STAGE 1: Data Collection

Hardware: Intel RealSense D415 stereo depth camera, ring-light illumination, matte black background.

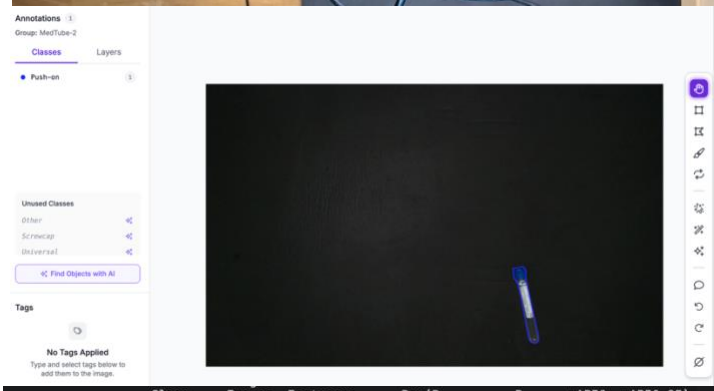
Protocol: 30 capture sessions, one tube per session, 100 frames each.

Output: 3,000 aligned RGB + 16-bit depth image pairs at 1280×720, 30 fps.

STAGE 2: Data Annotation

Platform: Roboflow (v4.0), upload, versioning, split management, export.

Method: SAM 2 (Segment Anything Model v2) click-based auto-segmentation followed by manual polygon refinement of cap boundaries.

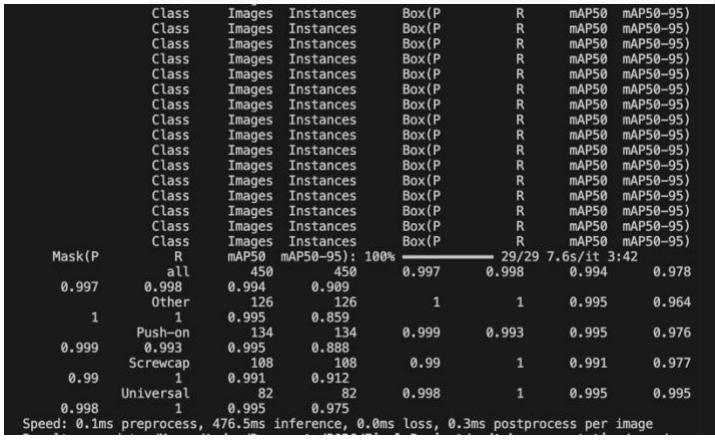


STAGE 3: Model Training

Inputs: RGB (3-channel), RGBD (4-channel), Depth-only (1-channel).

Hardware: M1 Max CPU and Roboflow cloud GPU.

Configuration: 100 epochs, image size 640×640, cosine LR schedule, aggressive augmentation.

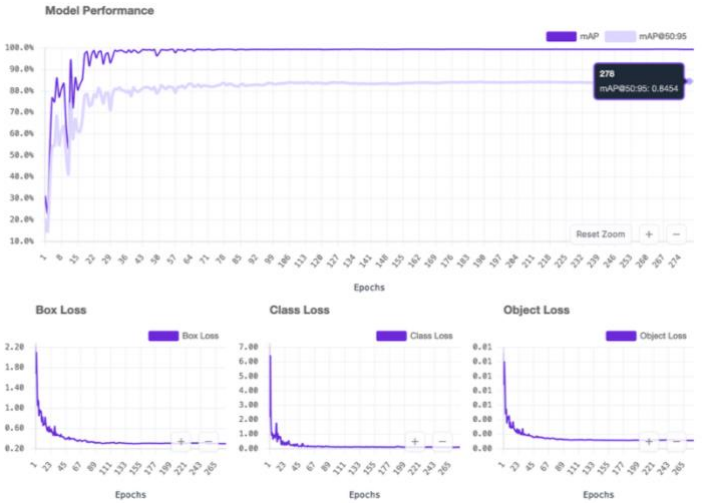


STAGE 4: Model Evaluation

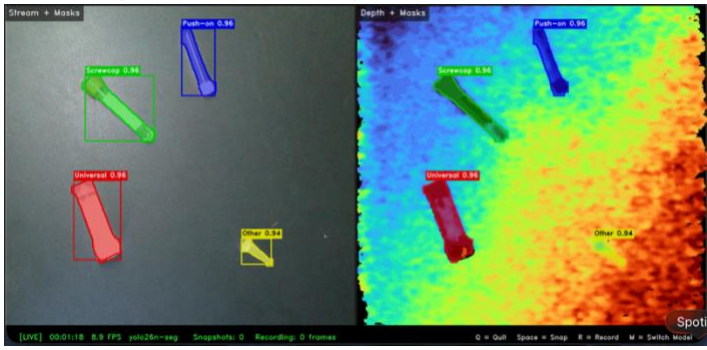
Metrics: COCO box and mask mAP at IoU 0.50 (mAP50) and 0.50–0.95 (mAP50-95).

Analysis: Per-class precision, recall, confusion matrices, F1 curves.

Speed: End-to-end inference latency measured on M1 Max CPU, averaged over the 450-image test split.



STAGE 5: Live Deployment

Display: 2×2 grid, raw RGB stream, RGB with masks, depth heatmap, depth heatmap with masks overlaid.

Features: Runtime model switching, snapshot capture, per-class colour coding.